Administration without UX: An AI-Based Computational Architecture for Automated Administrative Processing

Abstract

Public administrative systems are commonly implemented as digital services; however, their operational cores remain fundamentally dependent on human-operated interfaces such as authentication screens, menu navigation, structured input forms, and manual submission processes. These UX layers function primarily as adapters that translate human intentions into machine-readable states, rather than as intrinsic components of administrative logic. This paper proposes an AI-based administrative processing architecture that eliminates human-operated UX layers and reformulates public administration as a computational problem of state inference, rule evaluation, and consistency checking. Using South Korea's HomeTax system as a representative inter-domain case involving individuals, corporations, and the state, administrative procedures are decomposed into computational steps that exist solely to compensate for human operation. We demonstrate that a UX-less architecture can conceptually reduce processing steps, human-induced error propagation, and end-to-end administrative latency while preserving accountability through auditable computational models and human oversight mechanisms. This work reframes administration as an engineering problem rather than a usability problem and highlights the public value of removing human-operated UX layers from routine administrative processing.

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1. Introduction

Despite decades of digital transformation initiatives, modern public administrative systems remain fundamentally human-operated. Digital interfaces have replaced paper-based processes, yet the underlying assumption persists that individuals or corporate representatives must actively log in, navigate interfaces, interpret instructions, enter structured data, and submit declarations to trigger administrative processing. In such systems, UX components are often treated as core elements of administration itself, rather than as compensatory mechanisms necessitated by human limitations.

This human-centric design introduces several structural inefficiencies. First, administrative systems must devote significant computational and organizational resources to validating, correcting, and reconciling human input. Second, errors introduced at the point of data entry propagate through downstream processes, leading to delays, corrections, audits, and disputes. Third, the timing of administrative processing becomes dependent on when a human actor initiates interaction, rather than when a relevant real-world event occurs.

Recent advances in artificial intelligence—particularly in state inference, pattern recognition, and rule-based reasoning—enable administrative systems to interpret real-world states directly from authoritative data sources. This capability motivates a fundamental reconsideration of the role of UX in administration. Instead of optimizing UX to reduce human burden, it becomes possible to eliminate UX layers entirely for routine administrative processing and to treat administration as a computational pipeline.

This paper introduces the concept of "Administration without UX," in which human-operated interfaces are removed from routine administrative workflows. We argue that administration can be redefined as a process of inferring relevant states from available data, evaluating applicable legal and regulatory rules, and verifying internal consistency and accountability. By doing so, administrative systems can operate continuously and proactively, rather than reactively in response to human submissions.

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2. Related Work

Existing research on digital government and e-government has primarily focused on improving usability, accessibility, and service quality for human users. Studies in this domain emphasize interface design, user satisfaction, and the reduction of cognitive load through streamlined forms and guided workflows. While these efforts have yielded incremental improvements, they retain the fundamental assumption that humans must remain the primary operators of administrative systems. Another line of research explores AI-assisted decision support in the public sector. In these systems, AI tools are deployed to assist human officials or users by providing recommendations, detecting anomalies, or automating specific sub-tasks. However, these approaches typically position AI as an adjunct to existing UX-heavy workflows, rather than as a replacement for them. More recent work on data-driven government and algorithmic governance highlights the potential of automated processing based on integrated datasets. Yet even these frameworks often rely on human-triggered inputs or confirmations, particularly in domains involving taxation, benefits, and compliance.

In contrast to these approaches, the present study frames administrative inefficiency not as a problem of poor UX design or insufficient assistance, but as a structural consequence of human-operated interfaces themselves. By treating UX as an intermediate computational layer introduced to accommodate human participation, this work shifts the focus from improving UX to removing it. This architectural perspective distinguishes the proposed approach from prior studies and positions it as a contribution to the engineering of public administrative systems rather than to their usability optimization.

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3. UX-less Administrative Architecture

The proposed UX-less administrative architecture conceptualizes administration as a computational pipeline rather than a sequence of human interactions. At its core, the architecture consists of three primary functional components: state inference, rule evaluation, and consistency checking.

State inference involves determining relevant administrative facts from authoritative data sources. Rather than requiring individuals or corporations to declare their status through input forms, the system infers states such as income occurrence, employment status, or transaction completion directly from existing records.

Rule evaluation applies legal and regulatory logic to inferred states. This component encapsulates statutory rules in a machine-interpretable form, enabling systematic and reproducible application of regulations without human mediation.

Consistency checking ensures that inferred states and evaluated outcomes are internally coherent and auditable. This includes cross-validation across data sources, anomaly detection, and the generation of transparent audit trails.

The architecture is organized across three layers corresponding to individual administration, corporate administration, and inter-domain administration involving interactions among individuals, corporations, and the state. A unified AI interpretation engine operates across these layers, replacing UX-mediated data entry with direct computational interpretation. Human oversight is retained at defined checkpoints for dispute resolution, exception handling, and governance.

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4. Case Study: HomeTax

The South Korean HomeTax system provides a representative example of UX-heavy inter-domain administration. HomeTax integrates data from employers, financial institutions, corporations, and government agencies, yet still requires users to log in, navigate menus, fill out forms, and submit declarations to initiate processing.

From a computational perspective, many HomeTax procedures—such as authentication, form completion, submission confirmation, and correction—exist to translate human actions into machine-readable states. These steps introduce latency and error risks that are not intrinsic to tax administration itself.

By decomposing HomeTax workflows into computational functions, we identify which steps are essential to rule evaluation and which exist solely to accommodate human operation. This decomposition reveals that core tax logic can be executed independently of UX, provided that reliable state inference and verification mechanisms are in place.

5. Methodology

The evaluation methodology compares UX-driven and UX-less administrative architectures using three metrics: computational step count, probability of human-induced error propagation, and end-to-end processing latency.

Computational step count measures the number of discrete processing stages required to reach an administrative outcome. Human-operated workflows introduce additional steps related to authentication, navigation, input validation, and correction.

Human-induced error propagation captures the likelihood that errors originating from human input lead to downstream inconsistencies, reprocessing, or delays.

End-to-end latency measures the time from the occurrence of a real-world event to the completion of administrative processing. In UX-driven systems, latency is influenced by when humans initiate interaction, whereas UX-less systems can process events continuously.

The analysis relies exclusively on publicly documented workflows and does not use personal or confidential data.

6. Discussion

Eliminating UX layers shifts complexity from human procedures to computational models. While this increases the importance of accurate state inference and robust rule encoding, it also enables systematic auditing and reproducibility. Human oversight remains essential for governance, dispute resolution, and ethical accountability, but no longer serves as the primary driver of routine processing.

This shift has significant public value, including reduced administrative burden, fewer errors, and more timely processing. It also reframes administrative reform as an engineering challenge rather than a usability challenge.

7. Limitations

This study evaluates conceptual architectures rather than live deployments. Legal interpretation, policy enforcement, and societal acceptance are outside the scope of this analysis. Additionally, the feasibility of full UX removal depends on data availability, legal frameworks, and institutional readiness.

8. Conclusion

Administration without UX reframes public administration as a computational engineering problem. By eliminating human-operated UX layers, administrative systems can achieve greater efficiency, reliability, and transparency. The proposed architecture demonstrates the conceptual feasibility and public value of UX-less administration and provides a foundation for future research and implementation.